(iii) $\int x^3 e^{-x} dx.$

Applying Bernoulli's formula

$$\int u dv = uv - u'v_1 + u''v_2 - \cdots$$

$$\int x^3 e^{-x}\, dx = (x^3)(-e^{-x}) - (3x^2)(e^{-x}) + (6x)(-e^{-x}) - (6)(e^{-x}) + c$$

$$= -x^3 e^{-x} - 3x^2 e^{-x} - 6xe^{-x} - 6e^{-x} + c.$$

$$dv = e^{-x}dx$$

$u = x^3, \quad v = -e^{-x}$

$u' = 3x^2, \quad v_1 = +e^{-x}$

$u'' = 6x, \quad v_2 = -e^{-x}$

$u''' = 6, \quad v_3 = e^{-x}$

EXERCISE 11.7

Integrate the following with respect to x:

(1) (i) $9xe^{3x}$ (ii) $x \sin 3x$ (iii) $25xe^{-5x}$ (iv) $x \sec x \tan x$

(2) (i) $x \log x$ (ii) $27\ x^2 e^{3x}$ (iii) $x^2 \cos x$ (iv) $x^3 \sin x$

(3) (i) $\dfrac{x \sin^{-1} x}{\sqrt{1-x^2}}$ (ii) $x^5 e^{x^2}$ (iii) $\tan^{-1}\left(\dfrac{8x}{1-16x^2}\right)$ (iv) $\sin^{-1}\left(\dfrac{2x}{1+x^2}\right)$

11.7.8 Integrals of the form (i) $\int e^{ax} \sin bx\, dx$ (ii) $\int e^{ax} \cos bx\, dx$

The following examples illustrate that there are some integrals whose integration continues forever. Whenever we integrate function of the form $e^{ax} \cos bx$ or $e^{ax} \sin bx$, we have to apply the Integration by Parts rule twice to get the similar integral on both sides to solve.

Result 11.1

(i) $$\int e^{ax} \sin bx\, dx = \frac{e^{ax}}{a^2+b^2}[a \sin bx - b \cos bx] + c$$

(ii) $$\int e^{ax} \cos bx\, dx = \frac{e^{ax}}{a^2+b^2}[a \cos bx + b \sin bx] + c$$

Proof : (i)

$$\text{Let } I = \int e^{ax} \sin bx\, dx$$

$$\text{Take } u = \sin bx; du = b \cos bx\, dx$$

$$dv = e^{ax}; v = \frac{e^{ax}}{a},$$

Applying Integration by parts, we get

$$I = \frac{e^{ax}}{a} \sin bx - \int \frac{e^{ax}}{a} b \cos bx\, dx$$

$$I = \frac{e^{ax}}{a} \sin bx - \frac{b}{a} \int e^{ax} \cos bx\, dx$$

$$\text{Take } u = \cos bx; du = -b \sin bx\, dx,$$

$$dv = e^{ax}; v = \frac{e^{ax}}{a},$$

Again applying integration by parts, we get

$$I = \frac{e^{ax}}{a} \sin bx - \frac{b}{a}\left[\frac{e^{ax}}{a} \cos bx + \int \frac{e^{ax}}{a} b \sin bx\, dx\right]$$

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